Evidence-Guided Small-Part Sorting

A modular FANUC ER-4iA / Isaac Sim pipeline for camera-guided cube and screw sorting, with measured tests of neural perception and AI grasp ranking.

PROJECT AND MANUFACTURING PROBLEM

We built a modular FANUC ER-4iA / Isaac Sim pipeline that observes cubes and screws, estimates poses, generates feasible grasps, sorts parts, verifies outcomes, and records failures. Small-part sorting is sensitive to occlusion, calibration, dimensions, clutter, and grasp choice. Failures create retries, intervention, wasted time, and lost throughput.

OUR HYPOTHESIS AND NOVEL IDEA

Could AI not only detect objects, but also improve grasp selection? For every part, we generate several predefined, physically feasible grasps. A ranker selects the grasp least likely to fail. Success means acquiring the intended object, retaining it throughout relocation, and releasing it only at its destination. Geometry still rejects unreachable, poor-fit, or obstructed candidates; AI cannot make an unsafe grasp safe.

SCOPE AND ASSUMPTIONS

A small wall bounds the work area; parts are durable enough to survive drops and sufficiently non-bouncy not to escape it; part families, dimensions, and sorting destinations are known; destinations never fill or are emptied separately; cubes are upright; screws are isolated and lying down; and cameras have adequate visibility. Arbitrary piles, deformable parts, reflective finishes, and unrestricted moving arrivals remain outside the validated scope.

WHY YOLO11N AND COSMOS

We evaluated a fine-tuned YOLO11n because it was compact, fast, and our strongest tested neural detector in limited cube evaluation. We evaluated pretrained NVIDIA Cosmos3-Nano because it could rank structured feasible grasps without first collecting a large custom training set. Cosmos was not trained by us, and its confidence was not a calibrated success probability. Both were hypotheses, not predetermined winners.

CONSISTENCY OVER NOMINAL SPEED

We deliberately prioritize correct placement and grasp retention. A fast attempt has little manufacturing value when a drop triggers recovery, operator intervention, or downstream delay. The primary metric is therefore correct sorting; throughput is reported with failures and unresolved parts included.

MATCHED ISAAC SIM PILOT

Ten preregistered randomized scenes (1-10 objects) were replayed under four policies: 40 end-to-end runs and 55 object opportunities per policy. Lighting, materials, camera, sensor effects, object poses, and physics varied. All reports completed with zero missing or episode-level infrastructure failures.

Policy	Sorted	Scenes	/ sim min
DD	**45/55 (81.8%)**	**6/10**	**4.97**
DA	38/55 (69.1%)	3/10	3.76
YD	14/55 (25.5%)	1/10	4.28
YA	9/55 (16.4%)	1/10	3.09

DD = deterministic vision + deterministic grasp; **DA** = deterministic vision + guarded Cosmos; **YD** = YOLO-assisted vision + deterministic grasp; **YA** = both.

EVIDENCE-BASED DECISION

DD exceeded DA by 15.2 percentage points in matched object completion (paired 95% CI: 3.3-29.9 points). Cosmos changed the deterministic grasp 21 times; only 9 changed choices succeeded. The strict YOLO integration rejected an entire observation when one detection disagreed with geometry. We therefore retain DD instead of forcing AI into the deployed path.

SCALING TO NEW PARTS AND GRIPPERS

The goal is not universal out-of-box picking. A factory can repeat our process for a bounded new part family: add CAD/dimensions and symmetries, collect representative images, validate camera-to-robot error, define compatible grasps, collect outcomes, and train only if learning beats the baseline. Cubes plus geometrically different screws demonstrate more scalability than color alone. A new gripper changes finger geometry, stroke, TCP, collision envelopes, and candidate rules—not the entire pipeline.

PROJECT AUTOPSY

Midterms delayed meaningful work until roughly two weeks remained. Tasks were delegated too late. Non-lead contributors were not always given code-complete, bounded starting points, so limited ChatGPT usage was spent rebuilding shared foundations and sometimes exhausted before integration. Environment, credential, and dependency troubleshooting also consumed large portions of meeting time. We did not obtain a final physical-validation opportunity.

HOW WE WOULD IMPROVE

Start immediately after kickoff; schedule hardware access early; assign owned workstreams, interfaces, first deliverables, and tests at the beginning; have the lead prepare code-complete shared scaffolding so limited AI access is used on bounded work; hold a lead-run preliminary troubleshooting period before meetings; validate one camera-observed object end to end before expanding scope; and keep an AI component only when it beats a simple held-out baseline.

CSI EQUIPMENT AND LIMITATION

We used the organizer-provided FANUC digital twin, robot/gripper geometry, screw CAD, and Isaac Sim physics. We did **not** receive an opportunity to test the final pipeline on CSI hardware. Physical calibration, controller timing, grip forces, clearance, and sim-to-real reliability remain unverified; these results are simulation evidence only.

REPOSITORY AND REPRODUCIBLE RESULTS

github.com/ALL-Applied-AI-Network/msoe-ai-club-rockwell-automation-nvidia-hackath-new-team-8-b27f0a8f50/tree/experiments/randomized-four-combo-v1